ADITYA ENGINEERING COLLEGE

(An Autonomous Institution)

Approved by AICTE, Affiliated to JNTUK, Accredited by NBA & NAAC with 'A++' Grade

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Aditya Nagar, ADB Road, Surampalem, Kakinada District, A.P.

SlNo.:

PART - I

Programe : B - TECH

Hall Ticket No.: 21A91A0359

Name : CHESETTI SRINIVASA RAO

Examination :

Month-Year :

Branch : MECHANICAL ENGINEERING

Course Code : 201BS3T12

Course Name : Integral transforms & application of Partial Diffrom

Date of Exam : 18/11/2025

J. Pavan.

Signature of the Controller of Examinations

CH. Srinivasa Rao 18/11/25

Signature of the Student with date

18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	13

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART - 1 is torn / damaged, report to the invigilator and return the damaged booklet .
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras / Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Unit - V

10(a) Given data

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u \quad \text{where } u(x,0) = 6e^{-3x}$$

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u = 0$$

$$0 = u(0,y) = 8e^{-3y}$$

$$\frac{0 = u(0,y) = 8e^{-3y}}{\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial t} + u} = 0$$

$$u = (0,y) = \frac{\partial u}{\partial x} = 8e^{-3y}\, 4\left(\frac{\partial u}{\partial t}\right) + u = 0$$

$$\frac{u = (0,y) = \frac{\partial u}{\partial x}}{8e^{-3y}\, 4\left(\frac{\partial u}{\partial t}\right) + u} = 0$$

$$\frac{u = (0,y) = \partial u/\partial x - 8e^{-3y} 4\left(\frac{\partial u}{\partial t}\right) + u}{8e^{-3y} 4\left(\frac{\partial u}{\partial t}\right)^3 - u(0,y)\,\frac{\partial u}{\partial x}} = 0$$

$0 = 8e^{-3y} \quad 4\left(\frac{\partial u}{\partial y}\right)$

$4\left(\frac{\partial u}{\partial y}\right) = \frac{\partial u}{\partial t}\left(\frac{\partial^2 u}{\partial x^2}\right)$

$$\frac{\frac{\partial u}{\partial t}\left(\frac{\partial^2 u}{\partial x^2}\right)}{4\left(\frac{\partial u}{\partial y}\right)} = 0$$

let then the $\frac{\frac{\partial u}{\partial t}\left(\frac{\partial^2 u}{\partial x^2}\right)}{4\left(\frac{\partial u}{\partial y}\right)} = 0$, 10(b) Given Data

Boundary Conditions Equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$u(x,0) = 3 \sin n\pi x$, $u(0,t)$ ∮ $u(1,t) = 0$
Where $0 < x < 1$, $t > 0$

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} = 3 \sin n\pi x$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} = u(0,t) \text{ ∮ } u(1,t) = 0$$

$$\frac{\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}}{u(0,t) \text{ ∮ } u(1,t) = 0}$$

Then the
$$\frac{\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}}{u(0,t) \text{ ∮ } u(l,t) = 0}$$

Unit-I

1(a) Fourier Series

$$f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 3 \\ 6x & \text{if } 3 \leq x \leq 6 \end{cases}$$

$$f(x) = \begin{cases} x & 0 \leq x \leq 3 \\ 6x & 3 \leq x \leq 6 \end{cases} \quad \Rightarrow 0$$

$$f(x) = \begin{cases} x & \leq x \leq 3 \\ 6x & \leq x \leq 6 \end{cases}$$

$f(x) = \{ x \quad 0 \leq x \leq 3$

$f(x) = \{ 6-x \quad 3 \leq x \leq 6$

$f(x) = 6x \quad 3 \leq x \leq 6$

1 (b) Fourier Series $f(x) = x \cos x$ $(-\pi, \pi)$

$f(x) := x \cos x$ $(-\pi, \pi)$

$f(x) = x$

$x = \cos x$ $(-\pi, \pi)$

$f(x) = e^{x}$ $(0, 2\pi)$

$f(x) = x$ $(0, 2)$

$$f(x) = \frac{e^{x} (0, 2\pi)}{x\ (0,2)}$$

$$f(x) = \frac{e^{x} (\pi)}{x}$$

Unit-II

3(a) fourier integral.

$$\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2}\, d\lambda = \frac{\pi}{2} e^{-x}, x \geq 0$$

Given Data

$$\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2}$$

$$d\lambda = \frac{\pi}{2} e^{-x}, x=0$$

$$\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} \quad d\lambda = \frac{3.14}{2} e^{-x} \; x=0$$

$$\frac{\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2}}{d\lambda = \frac{3.14}{2} e^{-x} \cdot x} = x = 0$$

$$\frac{x = \int_0^\infty \frac{\cos \lambda x}{1+\lambda^2}}{d\lambda = \frac{3.14}{2} e^{-x}x} = 0$$

3 (b) Fourier sine & Cosine Series transform

$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 2 \\ 0 & \text{if } x \geq 2 \end{cases}$$

Given Data

$$f(x) = \begin{cases} 1 & 0 < x \leq 2 \\ 0 & x \geq 2 \end{cases}$$

$$f(x) = \{ 1 \quad 0 < x \leq 2$$

$$f(x) = \{ 0 \quad x \geq 2$$

$$f(x) = \begin{cases} 1 & 0 < x < 2 \\ 0 & x \geq 2 \end{cases}$$

$$f(x) = \begin{cases} \cos & 0 < x < 2 \\ \sin & x \geq 2 \end{cases}$$

Unit - III

6(a) Given that

$$L\{t^2 e^{-3t} \sin 2t\}.$$

$$L\{t^2 e^{-3t}\}\{\sin 2t\}$$

$$L\frac{\{t^2 e^{-3t}\}}{\{\sin 2t\}}$$

$$L = \frac{\{t^2 e^{-3t}\}}{\{\sin 2t\}} = 0$$

$$L = \frac{\{t^2 e^{3t}\}\{\sin 2t\}}{}$$

$$L = \sin 2t \; t^2 e^{3t}$$

6(b) Evaluate

Given Data

$$\int_0^\infty \frac{\cos 6t - \cos 4t}{t}$$

$$\frac{\int_0^\infty \cos 6t \int_0^\infty \cos 4t}{t}$$

$$\frac{\int_0^\infty \cos 6t}{t} = \frac{\int_0^\infty \cos 4t}{t}$$

$$\int_0^\infty \frac{\cos 4t}{t} = \int_0^\infty \frac{\cos 6t}{t}$$

$$\frac{\int_0^\infty \cos 4t}{\int_0^\infty \cos 6t}$$

$$\int_0^\infty \frac{\cos 2t}{t}$$

Unit IV

8)a Find $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$

Given Data

$$L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$$

$$L^{-1}\left\{\frac{s(s^2+a^2)}{(s^2+a^2)^2}\right\}$$

$$L^{-1}\left\{\frac{s(s^2+a^2)-(s^2+a^2)^2}{(s^2+a^2)^2}\right\}$$

$$L^{2}\left\{\frac{s(s^2+a^2)-(s^2+a^2)^2}{s+a}\right\}$$

$$L^{2}\left\{\frac{s}{s+a}\right\}$$

8(b) Laplace transform

Given Data

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + 3y = e^{-t}$$

$$y(0) = y'(0) = 1$$

$$\frac{d^2y}{dx} + e^{-t} = \frac{dy}{dx} + 3y = e^{-t}$$

$$\frac{\frac{d^2y}{dx} + e^{-t}}{\frac{dy}{dx} + 3y} = e^{-t}$$

$$dy + 3y = e^{-t}$$

$$\frac{dy + 3y}{e^{-t}}$$